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Demand side management of industrial electricity consumption: Promoting the use of renewable energy through real-time pricing

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HIGHLIGHTS

- Intermittent energy sources require demand side flexibility to minimise curtailment.
- Financial savings can be achieved concurrently with increased demand on wind energy.
- Peak-time demand reduction offers little benefit for wind generation.
- Price optimisation results in a fortuitous increase in wind energy consumption.
- Price optimisation of load provides benefits for consumers and grid operators alike.

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ABSTRACT

As the installed capacity of wind generation in Ireland continues to increase towards an overall goal of 40% of electricity from renewable sources by 2020, it is inevitable that the frequency of wind curtailment occurrences will increase. Using this otherwise discarded energy by strategically increasing demand at times that would otherwise require curtailment has the potential to reduce the installed capacity of wind required to meet the national 2020 target.

Considering two industrial electricity consumers, this study analyses the potential for the implementation of price based demand response by an industrial consumer to increase their proportional use of wind generated electricity by shifting their demand towards times of low prices. Results indicate that while curtailing during peak price times has little or no benefit in terms of wind energy consumption, demand shifting towards low price times is likely to increase a consumer's consumption of wind generation by approximately 5.8% for every 10% saved on the consumer's average unit price of electricity.

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1. Introduction

A future in which all our energy requirements are provided for using sustainable energy sources will require a strategic and coherent integration of technologies, old and new. However, it will also require a paradigm change in the manner in which people perceive energy. Modern society expects an unconstrained and reliable supply of cheap energy and this is the essence of the instantaneous functionality provided by today's electrical grids. However, many of the primary sources of renewable energy for electricity generation such as wind, wave, tidal, and solar are intermittent by nature and therefore, if unmanaged, cannot provide the same level of service that can be achieved by using conventional fully dispatchable fossil fuel generation. The flexibility lost by the lack of dispatchability of these energy sources must be replaced elsewhere in the supply/demand loop.

As Ireland pursues a target of 40% electricity generation from renewable sources by 2020, predominantly from wind, the need for non-generation based flexibility is becoming increasingly prevalent [1]. This is often achieved by trading with neighbouring electricity markets via interconnection or by capturing and storing generation when it is available using large scale energy storage systems such as pumped hydro energy storage (PHES) [2]. However, flexibility can also be achieved through the response of consumer's demand to system requirements using demand side management (DSM) which aims to change the amount or timing of a consumer's electrical demand. A study on the use of real-time pricing in Ireland's electricity market to influence the operation of domestic appliances indicated a resulting beneficial impact on the use of wind generated electricity [3]. This paper examines the impact that the load profile characteristics of two industrial electricity consumers has on their consumption of wind generated electricity as they respond to day-ahead real-time pricing signals in an attempt to minimise their operating costs.

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2. Background

2.1. Renewable electricity generation in Ireland

As with other European Union members, Ireland's strive towards a renewable future has been predominantly driven by EU legislation that mandates overall reductions of greenhouse gas emissions (GHGs) and the promotion of energy from renewable sources. Furthermore, Ireland's resilience against fluctuations in foreign fossil fuel markets has continued to weaken since the mid 1990's, hitting an all-time low in 2007 with imported fuels accounting for 91% of annual consumption [4]. These factors highlight the need for Ireland to exploit the island's indigenous renewable energy from wind, wave, tidal, and biomass sources. In response Ireland's electricity generation from renewable sources (RES-E) has increased from 2% in 1995 [5] to approximately 17.6% in 2011 [6] and is set to increase to upwards of 40% by 2020 [1]. Wind generated electricity has surpassed hydro power as the dominant contributor to Ireland's RES-E and is planned to contribute up to 86% of the 2020 target [1].

2.2. Curtailment of wind generation

Times of excessive wind generation require the curtailment of wind generators. In order to maintain frequency, transient and voltage stability a certain amount of conventional generation plant is required to remain online at any given time to provide grid stability. Currently this constrains instantaneous excursions of wind generation in excess of 50% in the combined market of the Republic of Ireland and Northern Ireland. Beyond this level of penetration wind generators are instructed to reduce output, the potential generation is wasted, and the potential revenue from the sale of that electricity is lost. A study commissioned by EirGrid (TSO) and published in June 2010 concluded that this wind penetration threshold could be extended to 60–80% of net load with minimal infrastructural upgrade requirements [7].

For installed wind capacities up to the point at which curtailment is required, the amount of conventional generation that can be displaced is primarily dependent on the capacity factor of wind in the region. For Ireland this is approximately 34% which means that installed wind generation will on average generate at 34% of its maximum export capacity. When the installed capacity of wind generation exceeds the limit of what can be managed without constraint, incremental gains in the amount of net demand that can be supplied by wind requires the addition of substantially increasing amounts of installed capacity. This diminishes the feasibility of incremental displacement of conventional generation plant by wind. This is demonstrated in Fig. 1 which depicts the impractical amounts of additional installed wind capacity required to make marginal gains in useable energy beyond the curtailment threshold.

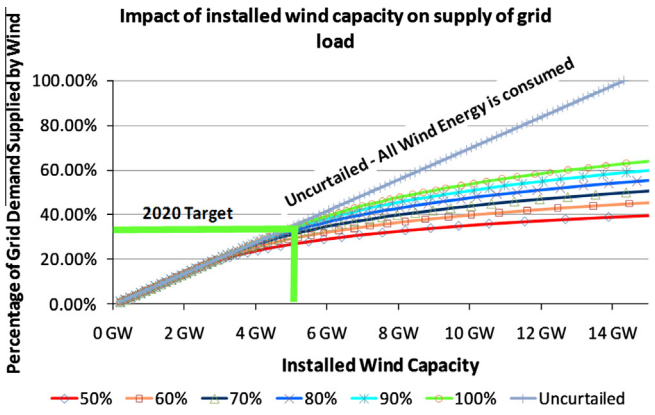


Fig. 1. Effect of various levels of curtailment on the feasibility of increasing penetration rates.

However, in order to maximise the realisable benefits of the available wind generation it will be necessary to increase the flexibility of load on the electrical grid using techniques such as DSM.

2.3. Demand side management

Wind generated electricity is inflexible as it cannot be truly dispatched or rescheduled due to the nature of its energy source. In order to increase the amount of conventional generation that can be replaced by wind it is necessary to achieve flexibility elsewhere. This has traditionally been considered the realm of large scale energy storage. In contrast to energy storage which aims to reallocate excessive supply to times of greater demand, Demand Side Management (DSM) aims to reallocate demand to times of increased wind generation. This is achieved through the use of dynamic load shifting. Conventional energy storage suffers from the inefficiencies inherited from the mechanical or chemical properties of the plant being employed. However, DSM has the potential to be 100% efficient as it does not require the conversion of energy to and from an intermediary form; instead it is the timing of the intended task that is displaced [8]. Schroeder and Modeling [9] suggests that centralised storage is more promising than DSM due to the complexity associated with a utility managing the shedding of distributed loads; however, here we are examining price based DSM in which loads passively react to a pricing signal rather than being actively dispatch by the utility. Although this approach avoids the need for centralised dispatch, it is necessary to ensure that the pricing signal encourages a response that is desirable to the utility. Nikonowicz et al. [10] suggests that distributed energy resources will become increasingly prevalent in the future, a concept that supports the approach used here. A number of studies have shown that DSM can indeed be used to shift demand to achieve financial savings for the customer [11–13]. The objective of this study is to examine whether such methods can not only achieve savings for the consumer but can simultaneously induce a response that facilitates the increased use of wind generation in the context of Ireland's electricity market.

A number of DSM initiatives are currently implemented by Ireland's TSO such as Short Term Active Response (STAR) [14,15], Powersave [16], Winter Peak Demand Reduction Scheme (WPDRS) [17,18], and Demand Bidding [19,20]. However, these strategies are limited to load shedding whereas in order to use DSM to provide the ancillary services needed to support the facilitation of wind energy, there is greater value in possessing an ability to increase demand as opposed to decrease demand as demonstrated in Paulus and Borggrefe [21] which examined the benefits of DSM in Germany's electricity market as wind penetration increases. As price based DSM encourages load shifting rather than load shedding, it offers the ability to strategically increase demand as well as decrease it. As communications technology continues to develop, it is becoming possible to succeed existing DSM initiatives with increasingly dynamic initiatives that enable higher levels of demand side participation in electricity markets and increase the demand side use of renewable energy sources as demonstrated in Bertoldi and Huld [22].

2.4. Ireland's electricity market

Similarly to other deregulated markets, Ireland's single electricity market (SEM) operates as a gross mandatory pool, into which all electricity generated or imported into Ireland must be sold and all wholesale electricity for consumption or export must be

purchased. Each trade day is comprised of 48 half hourly trading periods. Competitive bidding takes place one day ahead of delivery during which all dispatchable generators provide availability information as well as a number of dispatchable quantities and price pairs for the ensuing trade day. The prices submitted are expected to reflect the actual running cost of the generator unit and as such bid fluctuations are monitored to ensure that they correlate with the relevant fuel market prices. The availability of wind during each trade period determines the net load to be supplied by conventional generation plant. The spot market demand is cleared to meet predicted demand for each trading period and dispatch schedules are determined.

The clearing price is the price per MW h declared by the highest price generator required to meet demand. This determines the shadow price which will be awarded to all scheduled generators in a given trading period. This process is illustrated in Fig. 2.

When more expensive generators drive up the spot market price received by all generators, the margin made by less expensive generators is known as infamarginal rent. Generators are expected to recover start-up and no-load costs from their inframarginal rent. However, it is inevitable that the marginal generator during peak hours would not receive any inframarginal rent as no further generation will raise the price. To ensure that all generators at a minimum break even an additional pricing mechanism, 'Uplift' is added to the shadow price to produce the System Marginal Price (SMP) which is given to all generators. This generally results in severe price spikes during times of peak demand. An unconstrained schedule of predicted or "ex-ante" (EA) prices is then published at 16:00 h one day ahead of the trade date in question. Final or "ex-post" (EP) prices are published 4 days after the trade date after the inclusion of imbalances, constraints, and imperfections that had not been previously predicted during the ex-ante calculations.

Like the majority of the EU-27 member states, Ireland exercises explicit priority dispatch of renewables [23]. This is achieved by treating predicted wind availability as negative load during dispatch scheduling, resulting in all available wind power being accepted onto the grid. Consequently periods of high wind require the dispatch of fewer price-making conventional generation plants resulting in lower prices. This process lends itself to providing consumers with day-ahead, half-hourly pricing signals that are indicative of real-time RES-E penetration levels. This correlation will become more significant as the amount of wind generation increases. An electricity supplier tariff based on such signals would serve to facilitate the implementation of DSM by offering financial incentive to reallocate demand while simultaneously promoting the increased use of renewable energy. However, it is likely that the inclusion of 'Uplift' will reduce the potential benefits as it disproportionately distorts the relationship between the real-time price and the real-time demand being placed on the electricity system.

3. Methodology overview

The aim of this study is to determine if a correlation exists between the average price of electricity for an industrial electricity consumer on a real-time pricing tariff and their consumption of wind generated electricity.

The electricity consumption profiles of two medium voltage industrial electricity consumers with maximum import capacities of 1000 kV A or greater were observed for one year, from March 2011 to February 2012 inclusive, as part of this study.

- Consumer 1 is a highly efficient and modern cold storage warehouse with a highly flexible load profile shape.
- Consumer 2 is a manufacturing facility with reduced flexibility in the timing of their electricity consumption due to process constraints.

Both consumers avail of a real-time pricing tariff which reflects the half-hourly price of electricity in Ireland's electricity market were aware of the general daily trends of this tariff prior to participating.

During this study the participating consumers experienced varying levels of incentive to manipulate their demand profile. These incentives resulted in varying levels of financial savings for observation. These included seasonal influences on the real-time pricing tariff, a curtailment programme during winter months, and the implementation of a daily email information service by the electricity supplier. This service, which furnishes the consumers with day-ahead half-hourly pricing predictions as well as retrospective analysis of their electricity consumption with respect to

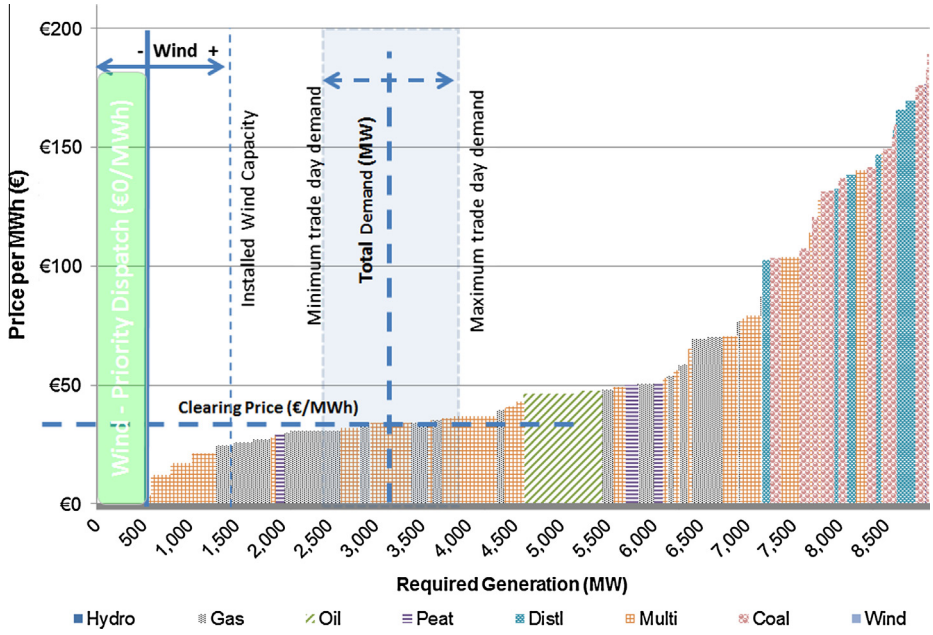


Fig. 2. Competitive bidding process illustrating clearing price, market spot demand and priority dispatch of wind.

the half-hourly price of electricity, is intended to encourage greater behavioural change among the electricity supplier's customer base, and will be discussed in more detail in Section 4.1.3. Due to commercial sensitivity issues, only post-analysis data will be presented in this paper.

3.1. Incentives for industrial electricity consumers

3.1.1. Real-time pricing

The participating electricity consumers are on a real-time pricing tariff which consists of all half-hourly wholesale electricity market prices plus a constant supplier mark-up which is charged in cents per kilowatt hour. The wholesale market price combines the SMP, as described in Section 1.4, transmission charges, distribution charges, market operator charges, as well as a number of levies and taxes. Overall, the SMP, which is most representative of the real-time demands being placed on the electricity system, accounts for approximately 60% of the total unit price. Fig. 3 depicts the average half-hourly prices during the period of this study as well as the average half-hourly prices during January and July. This chart clearly highlights the impact of the winter evening demand spike on prices which is no longer evident during summer months.

While significant variations in half-hourly electricity prices during winter months provide the consumers with substantial incentive to shift demand, this is greatly reduced during summer months when the potential to achieve savings by load shifting is limited by lower levels of variability. This is illustrated in Fig. 4 which analyses the standard deviation of daily half-hourly SMP prices, averaged for each month of the study. The analysis produces an oscillating profile which is representative of the opportunity to achieve savings from DSM throughout the course of the year.

3.1.2. Curtailment programme

The Winter Peak Demand Reduction Scheme (WPDRS), introduced in 2003, incentivizes commercial and small industrial electricity consumers to curtail demand during WPDRS hours, from 5 pm to 7 pm on business days during winter months [87,110]. For each of the four winter months from November to February inclusive, a baseline peak-time consumption quantity is established based on the consumer's demand between 5 pm and 7 pm during the previous 3 months. Participants then nominate a reduction commitment in respect of this baseline. The participant receives a 'profile payment' in reward for the energy saved and a further 'reliability payment' in reward for reliably providing DR services to the grid. Failure to adhere to the reduction commitment results in a 'reliability charge' for the participant. As an example, during the winter of 2011/2012 these payments and charges were €100/MWh, €224/MWh, and €783/MWh respectively [111]. In order to avoid reliability charges, a participant can adjust their reduction commitment quantity up to 12 pm on a given business day.

While both consumers partook in the WPDRS scheme, Consumer 1 had been avoiding marginal increases in electricity prices between 5 pm and 7 pm during the previous 3 months, reducing their baseline and diminishing their potential gains from the scheme. Consumer 2, on the other hand, had a high baseline which provided them with a substantial incentive to reduce demand during WPDRS hours.

3.1.3. Daily performance reporting

During the final 3 months of this study the consumers' electricity supplier provided a daily email information service which furnished the consumers with day-ahead half-hourly pricing predictions as well as retrospective analysis of their electricity consumption with respect to the half-hourly price of electricity. The daily email consists of five charts that are used by the consumers to inform on-going decisions regarding their use of electricity.

The chart in Fig. 5 provides the consumer with a prediction of the unit price of electricity during each half hour of the following day. This is based on a prediction that is published daily by Ireland's single electricity market operator (SEM-O). However, due to errors in demand forecasting, wind forecasting, and generator availability etc. this prediction is not always 100% accurate. In order to inform consumers of the potential inaccuracy in the prediction, a standard deviation of the positive and negative errors for each respective half hour during the previous 28 days is represented using the red shaded area surrounding the predicted price. Narrow bands in the shaded area indicate times during which predictions have been accurate while wide bands indicate times of substantial volatility in the prediction error.

The actual unit prices that transpired are published by SEM-O 4 days retrospectively once the market settlement has been finalised. Once this occurs, this information in presented to consumers alongside the price that had been predicted for the day as shown in

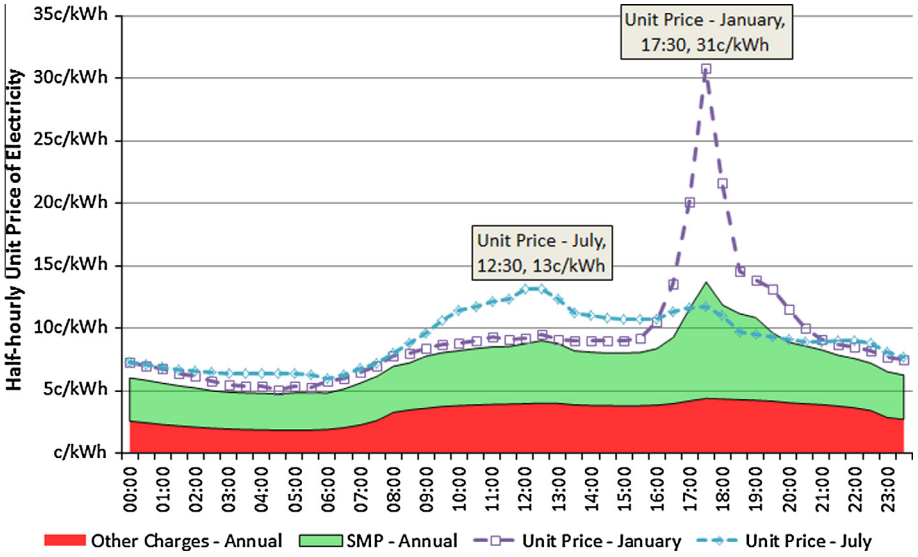


Fig. 3. Annual average half-hourly unit price of electricity separated into SMP and other charges compared to average half hourly prices for January (winter) and July (Summer).

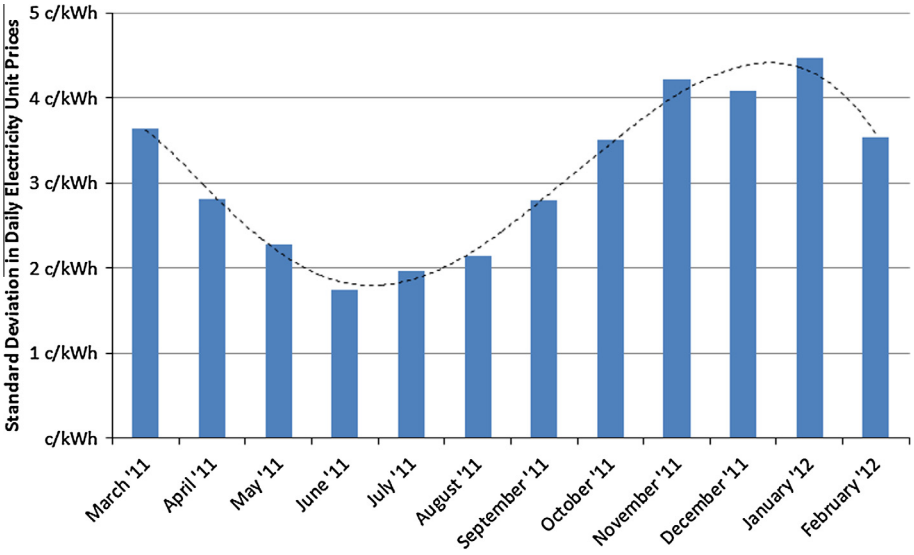


Fig. 4. Monthly averaged daily standard deviation of half-hourly electricity prices (SMP component).

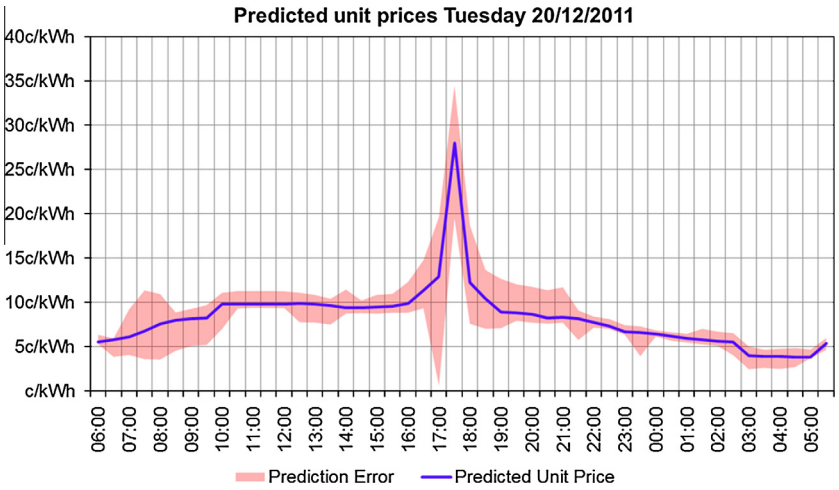


Fig. 5. Day-ahead unit price predictions including a prediction error analysis.

Fig. 6. This is intended to educate the consumer on the behaviour of actual prices versus predicted prices and to provide a level of certainty in the reliability of the price predictions.

The chart in Fig. 7 shows the consumer's quarter-hourly demand during the more recent day for which actual price and consumption data is available. It also depicts the consumer's hourly expenditure on electricity during that day. This chart provides the consumer with a tangible link to their daily activity and highlights the impact that price spikes have on their electricity spend in a manner that is intended to encourage the dispersion of demand from high price times.

The chart in Fig. 8 allows the consumer to analyse their daily electricity consumption and resulting expenditure during the previous 30 days. This is primarily used to illustrate the impact of variations in consumption quantities on cost.

The aim of DR, from the point of view of the consumer, is to reduce the average unit price (AUP) of their electricity by using more power during times of low prices and less power during times of high prices. The chart in Fig. 9 compares the consumer's average unit price based on their actual half-hourly consumption, known as their load weighted average (LWA) AUP, to the average unit price if they had a constant unit load. This chart enables the consumer to visualise trends in the wholesale price of electricity as well as their own AUP.

The customer's LWA AUP depicted in Fig. 9 is not only influenced by the consumer's DR performance but is also influenced fluctuations in the daily wholesale price of electricity. Furthermore, the consumer's 30 day demand and expenditure depicted in Fig. 8 is influenced by DR performance, fluctuations in the daily wholesale price of electricity, and variations in the consumer's demand. While other valuable information is presented in these charts, it is difficult for the consumer to isolate the impact of their DR performance on their AUP. As such it is necessary to provide the consumer with a metric that illustrates the consumer's DR performance by isolating the impact of the consumer's load profile shape on their AUP and ignoring other factors such as fluctuations in the daily wholesale price of electricity and variations in the consumer's demand. To do this, a unit load profile is used as a yard stick against which changes in the consumer's profile can be benchmarked. The chart presented in Fig. 10 demonstrates the difference in AUP achieved by the consumer's load profile when compared to the flat load benchmark on a daily basis, calculated using a metric described in Section 1.7. For example, a data point of -5% means that the consumer's AUP is 5% lower than it would be if they had

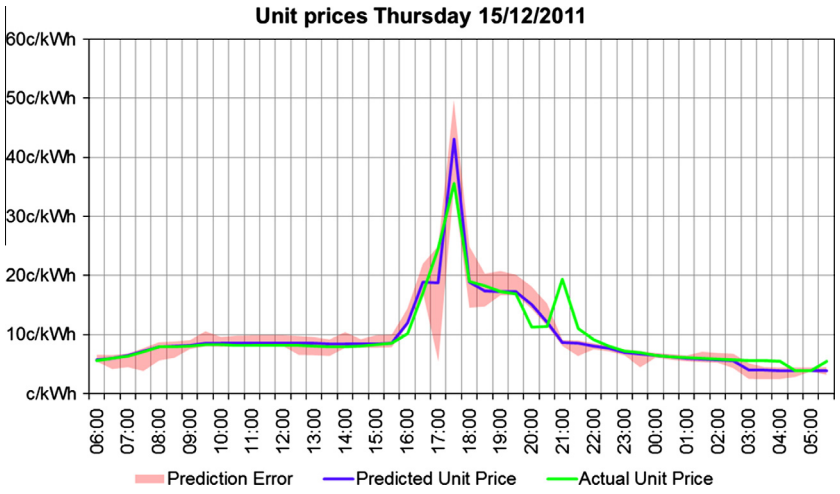


Fig. 6. Actual unit prices compared to predicted unit prices from 4 days previously.

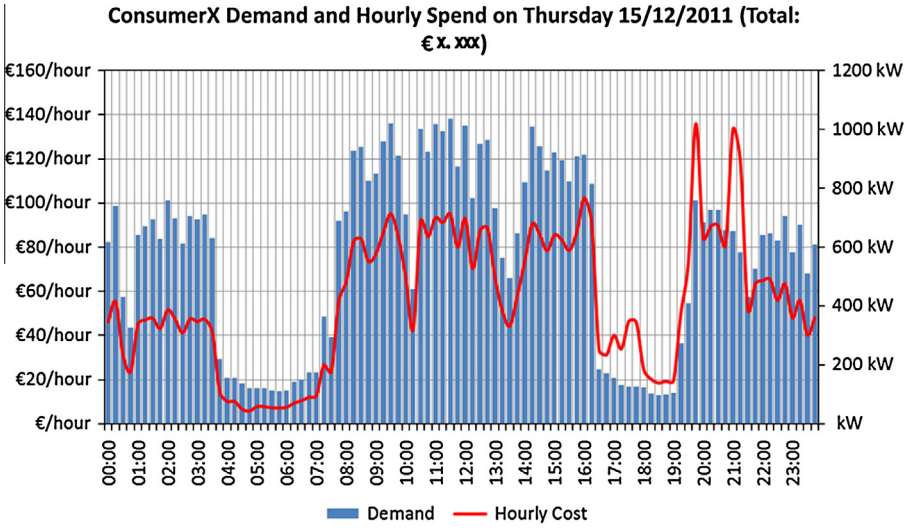


Fig. 7. Quarter-hourly electricity demand and hourly cost from 4 days previously.

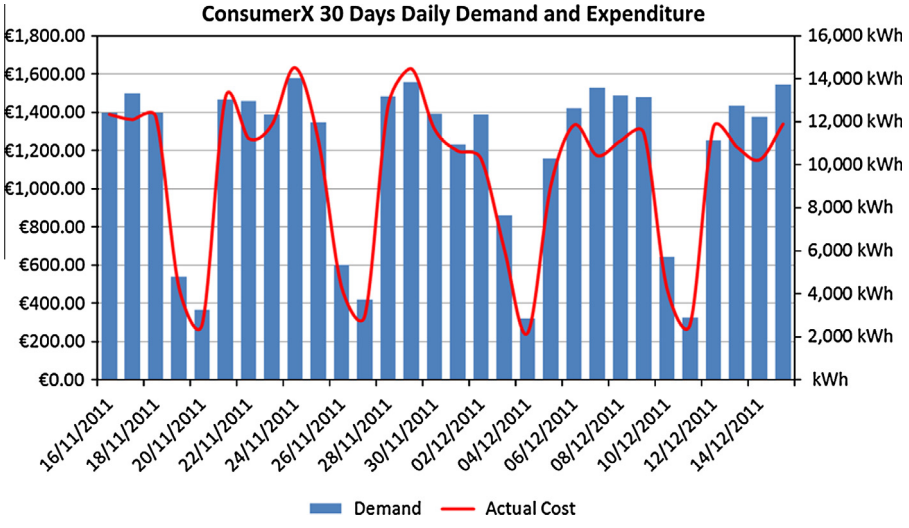


Fig. 8. Daily electricity consumption and cost for the previous 30 days.

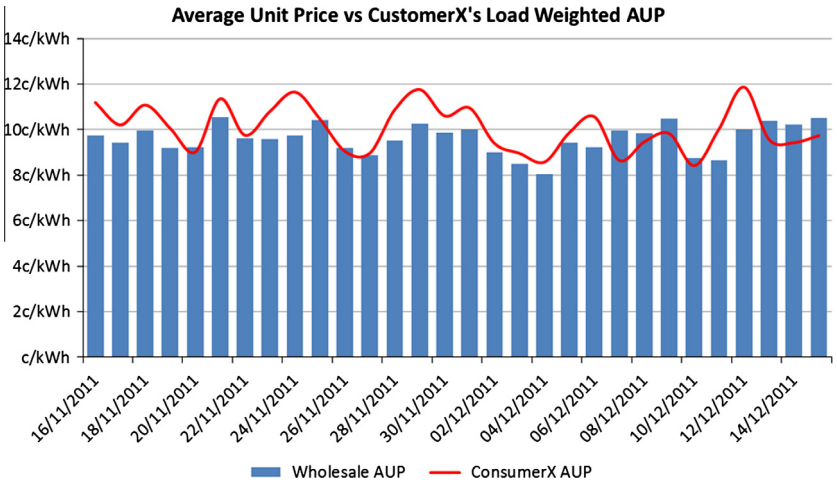


Fig. 9. Daily average unit price of electricity for a unit load compared to the consumer's load weighted average unit price during the previous 30 days.

a constant demand throughout the day. Furthermore, the delta between datapoints indicates the savings due to DR performance between two individual days. For example, if two consecutive days have values of +3% and −5% respectively, this indicates that the consumer saved 8% on their cost of electricity due to DR performance. This allows the consumer to monitor the progress and benefits of their DR initiatives.

3.2. Price and wind usage comparison metrics

In order to assess a consumer's DR performance in terms of cost savings and wind energy usage in a manner that independent of general fluctuations in wholesale prices, the consumer's demand, and wind availability, it is necessary to benchmark the consumer against an analogue that will be equally impacted by these factors. In the case of price, the consumer's performance is expressed as a percentage variation between the consumer's daily AUP, which is impacted by variations in their consumption throughout the day, and a benchmark AUP which is based on the consumer having a constant demand throughout the day. This is calculated for each day using Eq. (1). Example results from this calculation are illustrated in Fig. 10.

$$Performance_{Price} = \frac{Avg_Price_{Unit_Demand}}{Avg_Price_{Consumer_Demand}} * 100 \quad (1)$$

where

$$Avg_Price_{Unit_Demand} = \frac{\sum_{i=1}^{48} Price(i)}{48} \quad (2)$$

where $Price(i)$ is the unit price of electricity in a given half hour

$$Avg_Price_{Consumer_Demand} = \frac{\sum_{i=1}^{48} Demand_{Consumer}(i) * Price(i)}{\sum_{i=1}^{48} Demand_{Consumer}(i)} \quad (3)$$

where $Demand_{Consumer}(i)$ is the consumer's electricity demand in a given half hour and $Price(i)$ is the unit price of electricity in a given half hour.

In the case of wind, the consumer's daily performance is expressed as a percentage variation between the percentage of the consumer's electricity that was derived from wind generation and the percentage of national demand that was derived from wind generation. This is calculated for each day of the observation period using the following equation:

$$Performance_{Wind} = \frac{Wind_{Consumer}}{Wind_{National}} * 100 \quad (4)$$

where

$$Wind_{Consumer} = \frac{\sum_{i=1}^{48} Demand_{Consumer}(i) * \frac{Generation_{Wind}(i)}{Demand_{National}(i)}}{\sum_{i=1}^{48} Demand_{Consumer}(i)} * 100 \quad (5)$$

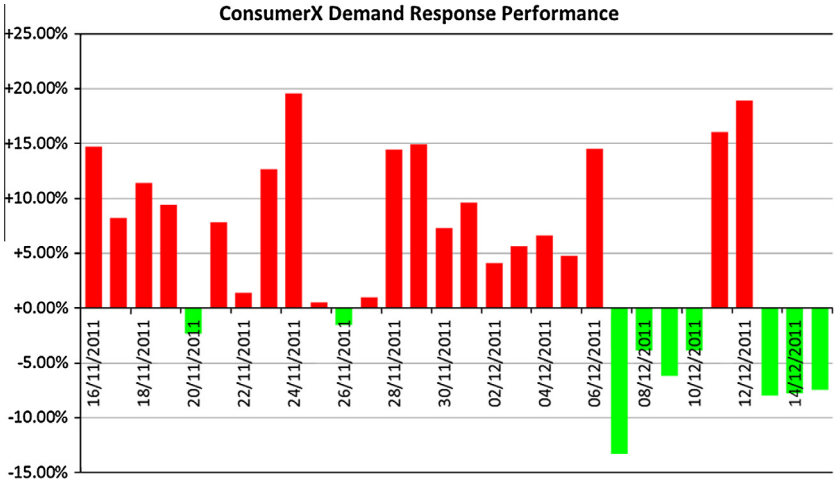


Fig. 10. Demand response performance during the previous 30 days.

where $Demand_{Consumer}(i)$ is the consumer's electricity demand in a given half hour, $Generation_{Wind}(i)$ is the quantity of electricity generated using wind in the Republic of Ireland in a given half hour expressed in Megawatt hours and $Demand_{National}(i)$ is the quantity of electricity consumed in the Republic of Ireland in a given half hour expressed in Megawatt hours

$$Wind_{National} = \sum_{i=1}^{48} \frac{Generation_{Wind}(i)}{Demand_{National(i)}} * 100 \quad (6)$$

where $Generation_{Wind}(i)$ is the quantity of electricity generated using wind in the Republic of Ireland in a given half hour expressed in Megawatt hours and $Demand_{National}(i)$ is the quantity of electricity consumed in the Republic of Ireland in a given half hour expressed in Megawatt hours

The aforementioned equations are used to assess the participating consumers' respective performances during the observation period in order to analyse whether or not a positive relationship exists between price performance and wind performance. It should be noted that when reading price performance, low and negative values are desirable as they indicate that the consumer's profile led to a lower AUP when compared to the benchmark whereas high and positive values are desirable in relation to wind performance as this indicates that the consumer used a higher percentage of wind generated electricity when compared to the benchmark.

4. Results and discussion

The participating consumers differ substantially in terms of industry type, demand profile flexibility, and levels of active response during the observation period. This is partially demonstrated in Fig. 11 which illustrates that consumer1's demand profile resulted in a consistently good price performance whereas consumer2's demand profile consistently resulted in a much poorer price performance by comparison. This section presents and discusses the results for the individual participants before considering the hypothesis under investigation in the context of the combined findings from the two case studies.

4.1. Consumer 1 results and discussion

As a highly efficient cold storage facility, consumer1 demonstrated a very flexible load profile. While the plant operation was not automatically rescheduled in response to day-ahead pricing predictions, the operations schedule was regularly updated by management during the observation period in response to prevailing daily pricing trends. This primarily involved shifting as much discretionary demand as possible to night time hours and limiting necessary but time-independent day time demand to hours of reliably low prices times as advised by the electricity supplier. As such, a very favourable price performance was achieved on a monthly basis as demonstrated in Fig. 12.

While consistent efforts were made to maximise demand response performance throughout the year, the data depicts an evident seasonality in the price performance levels that are achieved. This result had been expected due to the reduced volatility of prices during summer months when compared to winter months as discussed in Section 4.1.1. The monthly price performances observed closely correspond with the data in Fig. 4 which depicts the varying levels of opportunity for cost savings from DR on a monthly basis.

Importantly, an inversely proportional relationship between wind performance and price performance is illustrated by the reflective nature of the trends plotted in Fig. 12. This provides an indication that a consumer's efforts to reduce costs by using DR in response to a pricing signal will fortuitously increase the proportion of wind generation they use.

4.2. Consumer 2 results and discussion

Consumer2, a manufacturing plant, initially exhibited high demand during high priced times, low demand during low priced times, and was thought to have a very inflexible load profile due to the direct impact that load pattern alterations would have on work shifts. Consumer2's monthly wind performance and price performance are compared in Fig. 13. As with Consumer1, this demonstrates an inversely proportional relationship between price performance and wind performance.

For the first 9 months of the observation period, consumer2 maintained its legacy operations schedule which, as illustrated in Fig. 13, resulted in a poor price performance and below average consumption of wind generation. Despite significant incentive to shift demand from peak-hours during the first month of WPDRS in November, as discussed in Section 4.1.2, no noticeable improvement in their price performance is observed until December when the daily energy performance reporting service was implemented. Once implemented, this service provided the information needed to inform business decisions related to load shifting. While initial efforts to curtail peak-time demand were focused on maximising

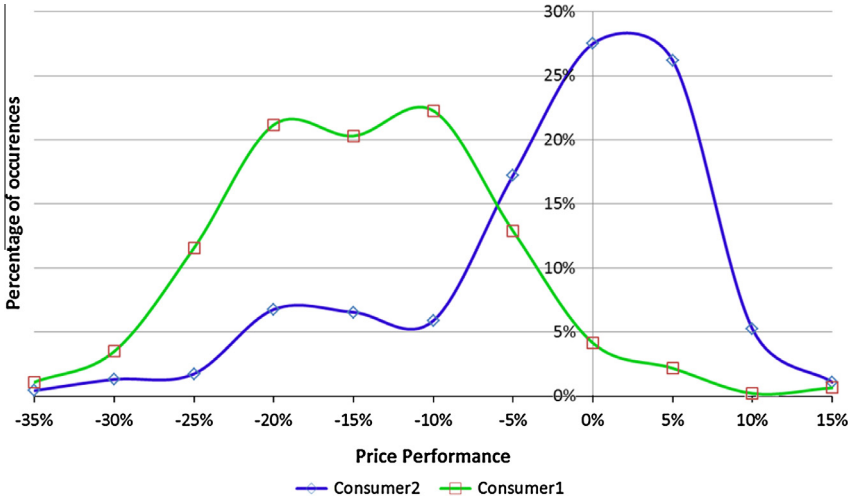


Fig. 11. Histogram of the participating consumers' price performance during the observation period.

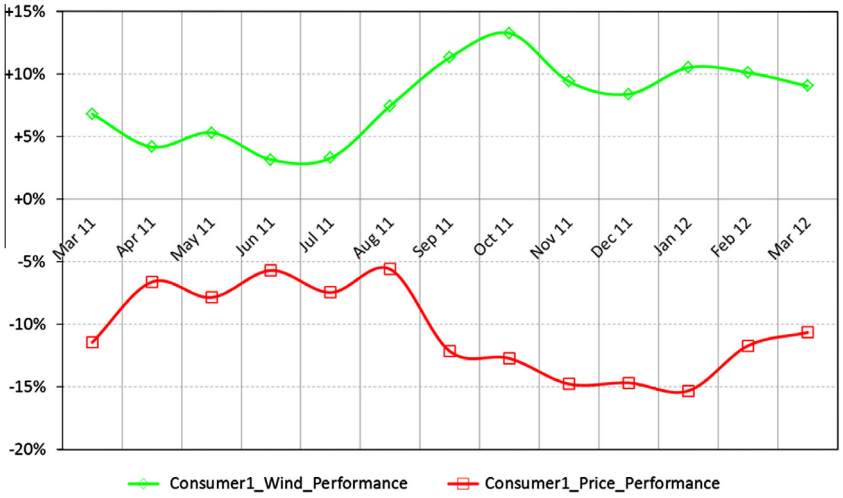


Fig. 12. Consumer1's monthly wind performance and price performance during the observation period.

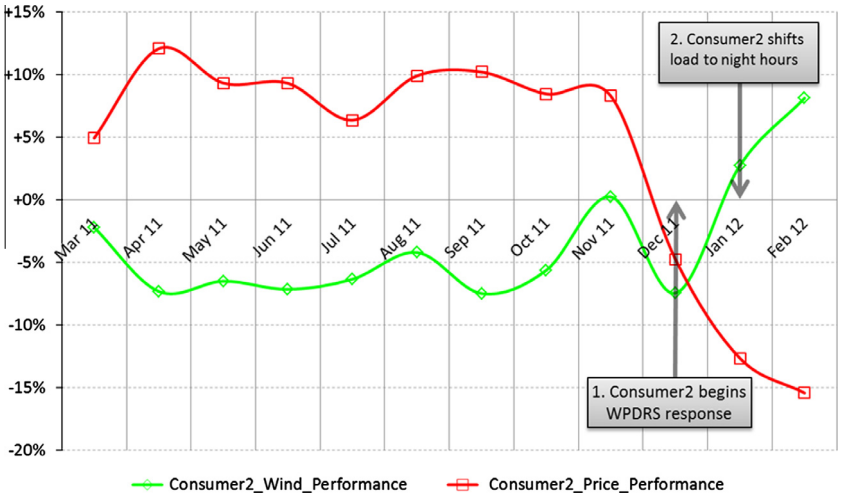


Fig. 13. Consumer2's monthly wind performance and price performance during the observation period.

returns from WPDRS, when the added benefit of the reduced AUP was observed during December, as highlighted by the price performance in Fig. 13, consumer2 implemented a substantial change to working shifts. Beginning in January, the highest energy intensive processes were scheduled during night time hours, medium intensity processes were scheduled during low priced day time hours, and minimal activity was scheduled during high priced day time hours.

Although initial price performance improvements resulting from peak-time curtailment were substantial, there was no improvement in wind performance in December. This outcome had been expected as the peak-time price spike is disproportionately impacted by the 'uplift' component of the price, described in Section 3.4, which reduces the correlation between wholesale prices and real-time system generation requirements. However, wind performance greatly improves following the implementation of load shifting involving strategic demand growth during night time hours. This concurs with the findings of Paulus and Borggrefe (2011) [21] which concludes that an ability to curtail demand offers little benefit to the support of wind generation, rather the greatest value offered by DR to the facilitation of wind generation is derived from an ability to strategically increase demand. Continued improvements in price performance in February are seen to correspond with increased consumption of wind generation.

4.3. Concluding results and discussion

Preliminary indications from the data presented in Sections 5.1 and 5.2 have suggested an inversely proportional relationship between price performance and wind performance when a consumer alters their demand profile in response to a pricing signal that is representative of the wholesale cost of electricity. The chart in Fig. 14 shows the average daily wind performance for various levels of daily price performance for consumers 1 and 2. Both of these trends provide further evidence that wind performance increases as a result of increases in price performance. The dashed line represents a linear estimation of the two primary trends.

The slope of the linear trend line estimates that for every 10% reduction in the AUP due to price incentivised DR, the proportion of the consumer's demand derived from wind generation will increase by 5.8%, providing benefits for both the consumer and the electricity system. Similarly, a 10% increase in AUP due to poor price performance will reduce the proportion of wind generation by 5.8%.

The consumer's examined here are industrial electricity consumers and so are metered using half-hourly interval metering which enabled the use of a half-hourly electricity tariff. Such a tariff has the ability to strategically incentivise any given half hour, hence directing demand towards that time and potentially

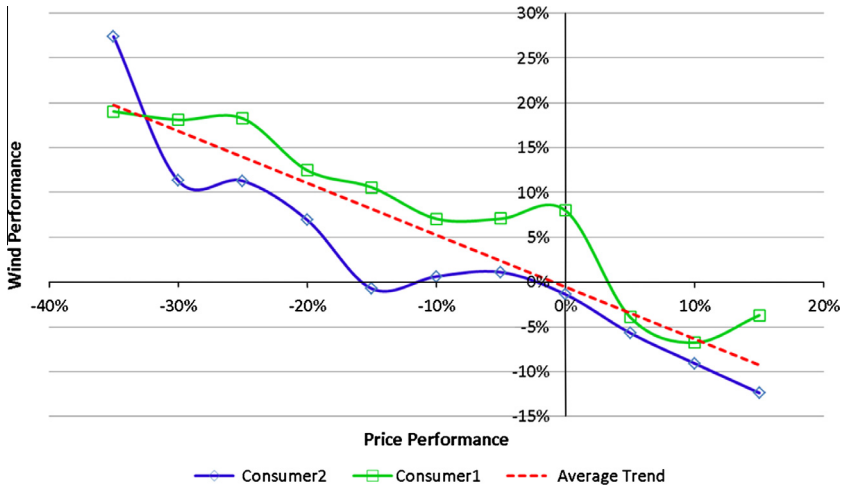


Fig. 14. Average wind performance mapped to various levels of price performance for Consumers 1 and 2 including a linear estimation of the two trends combined.

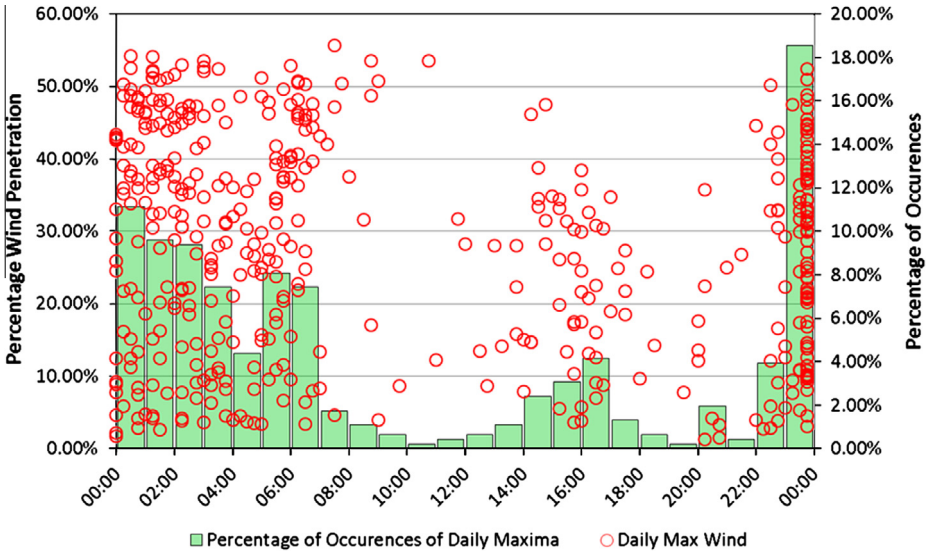


Fig. 15. Timing and magnitude of daily wind penetration maxima during the observation period (left axis) and a histogram showing the frequency of daily maxima during each hour (right axis).

reducing the requirement to curtail wind generation during times of high penetration and otherwise low demand. The chart in Fig. 15 shows that the majority of daily wind penetration maxima occur during night time hours. Furthermore, day time wind penetration maxima are not only rarer but also generally lesser in magnitude. With 76% of daily wind penetration maxima occurring between 11 pm and 8 am as a result of low system demand during that time, standard day/night rate electricity tariffs could potentially offer a reasonable proportion of the benefits demonstrated in this study. However, the shifted demand would be less focused, occurring over a longer time horizon, and of lower magnitude during a given half hour.

5. Conclusions

This study analysed the potential for the implementation of price based demand response by an industrial consumer to increase their proportional use of wind generated electricity. The foundations for the hypothesis are founded on the impact of wind generation on wholesale electricity prices in Ireland's electricity market due to the priority dispatch of the energy source. The study examined the consumption data of two industrial consumers, both of which use a half-hourly electricity tariff that is based on the wholesale price of electricity. Both case studies demonstrate a clear correlation between reductions in the consumers' AUP and increases in their consumption of wind generated electricity. While little or no increase in wind consumption was achieved by solely curtailing in response to peak price times, shifting demand to low price times is shown to provide substantial benefits. It was found that a 10% reduction in a consumer's AUP due to the implementation of price incentivized DR typically results in an increase in consumption of wind generation of approximately 5.8%. This provides an incentive for consumer's to participate that will result in a fortuitous benefit for the national electricity system.

While a half-hourly electricity tariff and interval metering would ideally be used for such a DR programme, in the absence of such technology a standard day/night meter and electricity tariff offers the potential to capture a reasonable portion of the potential gains.

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